

Phase 3 Technical Design – Tenant Model

Status: APPROVED & FROZEN

Phase: Phase 3 Only

Depends On: Phase 1 Foundation & Project Setup, Phase 2 Authentication

Architecture Baseline: Product & Architecture Requirements v1.0

1. Purpose

Phase 3 introduces the Tenant Model.

A tenant represents an independent business operating within the platform.

Examples include:

- Barbers
- Hair salons
- Nail artists
- Clinics
- Therapists
- Coaches
- Personal trainers
- Consultants
- Other appointment-based businesses

The purpose of this phase is to establish the business ownership boundary of the platform.

This phase defines:

- What a tenant is
- How a tenant is created
- How tenant ownership is established
- How tenant lifecycle is managed
- How tenant data ownership works
- How tenant records are maintained

This phase intentionally does not introduce memberships, permissions, roles, services, staff, appointments, branding, notifications, or platform administration.

2. Scope

Included

Phase 3 includes:

- Tenant concept
 - Tenant ownership model
 - Relationship between users and tenants
 - Tenant lifecycle
 - Tenant status model
 - Tenant creation rules
 - Tenant update rules
 - Tenant deactivation rules
 - Data ownership principles
 - High-level persistence design
 - Security considerations
 - Testing strategy
 - Acceptance criteria
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Excluded

The following are explicitly out of scope:

- Tenant memberships
- Tenant switching
- Tenant roles
- Permission system
- Staff management
- Client management
- Services
- Service categories
- Schedules
- Availability
- Appointments
- Branding
- Notifications
- Platform administration
- Suspension workflows

These capabilities belong to future phases.

3. Objectives

The objectives of Phase 3 are:

1. Introduce tenants as first-class business entities.
 2. Establish tenants as the ownership boundary of business data.
 3. Allow authenticated users to create tenants.
 4. Associate each tenant with an initial owner reference.
 5. Introduce a tenant lifecycle model.
 6. Support tenant updates.
 7. Support tenant deactivation.
 8. Preserve future extensibility without implementing future functionality.
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4. Tenant Concept

A tenant represents one independent business.

Examples:

- Ahmad Barber
- Yousef Barber
- Luxury Nails
- Prime Therapy Clinic
- Fit Coach Studio

A tenant is the root business entity within the platform.

Future business entities will belong to a tenant.

Examples of future tenant-owned entities include:

- Services
- Staff profiles
- Client profiles
- Schedules
- Availability rules
- Appointments
- Invitations
- Notifications
- Branding settings
- Audit records

These entities are not implemented in Phase 3.

5. Tenant Ownership Model

Every tenant must have:

- A creator
- An initial owner reference
- A lifecycle status
- Business identity metadata

The initial owner reference identifies the user who created and initially owns the tenant.

Only authenticated users may create tenants.

Unauthenticated users may not create tenants.

6. Ownership Transition Rule

The owner reference introduced in Phase 3 is a bootstrap ownership mechanism.

When Phase 4 introduces tenant memberships, the initial owner must also exist as an OWNER membership.

Tenant access control must ultimately derive from memberships rather than directly from the owner reference.

The owner reference may remain for auditing and convenience but should not become the long-term authorization source of truth.

This rule exists to ensure that the platform evolves toward membership-based authorization while preserving historical ownership information.

7. Relationship Between Users and Tenants

Phase 2 introduced global users.

Phase 3 introduces tenants.

The relationship established in Phase 3 is:

User creates Tenant
Tenant stores initial owner reference

This relationship exists only to bootstrap tenant ownership.

It is not the final authorization model.

Phase 3 must not implement:

- Membership records
- Access lists
- Role assignments
- Tenant switching
- Permission checks

These capabilities are deferred to later phases.

8. Tenant Lifecycle

The approved tenant lifecycle consists of:

- PENDING_ONBOARDING
- ACTIVE
- DEACTIVATED

No additional lifecycle states exist in Phase 3.

Platform-level suspension functionality is deferred to a future phase that introduces platform administration capabilities.

9. Tenant Status Model

PENDING_ONBOARDING

The tenant exists but has not completed future onboarding requirements.

All newly created tenants begin in this status.

ACTIVE

The tenant is considered operational.

Future onboarding phases may transition tenants into this state.

Phase 3 does not implement onboarding logic.

DEACTIVATED

The tenant has been deactivated.

The tenant record remains preserved.

Historical information remains intact.

The tenant is no longer considered operational.

10. Tenant Creation Rules

Tenant creation requires an authenticated user.

A tenant should contain basic business identity information.

Examples of business identity information include:

- Business name
 - Business category or type
 - Timezone
 - Initial owner reference
 - Lifecycle status
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Business Name Rules

Business name must:

- Be provided
- Be human-readable
- Not be empty
- Not consist only of whitespace

Examples:

Valid:

- Ahmad Barber
- Luxury Nails

- Prime Therapy Clinic

Invalid:

- Empty value
 - Whitespace-only value
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Business Type Rules

The tenant should identify its business type.

Examples:

- Barber
- Salon
- Nail Artist
- Therapist
- Clinic
- Coach
- Personal Trainer
- Consultant
- Other

Business type exists for categorization purposes.

Business type must not affect authorization, security, or tenant isolation.

Timezone Rules

The tenant should store a timezone.

Future scheduling and booking systems will rely on tenant timezone information.

Phase 3 only stores this information.

Phase 3 does not implement scheduling behavior.

11. Tenant Update Rules

Tenant updates are limited to tenant metadata.

Examples:

- Business name
- Business type
- Timezone

Updates must preserve tenant identity.

Updates must not:

- Change ownership unexpectedly
- Create memberships
- Assign roles
- Create services
- Trigger onboarding workflows

Phase 3 only supports metadata maintenance.

12. Tenant Deactivation Rules

Tenant deactivation is a lifecycle transition.

Deactivation is not deletion.

A deactivated tenant:

- Remains stored
- Retains historical information
- Retains ownership information
- May potentially be reactivated in the future

Phase 3 does not define reactivation workflows.

Deletion Policy

Hard deletion is not part of the Phase 3 tenant model.

Tenant records should be preserved.

This supports:

- Historical integrity
- Future auditing
- Future support workflows

13. Data Ownership Principles

The tenant is the ownership boundary of business data.

Every future business entity must belong to exactly one tenant.

Examples include:

- Services
- Staff profiles
- Client profiles
- Appointments
- Availability rules
- Branding settings
- Notifications
- Audit records

Resources are not shared between tenants in the MVP.

A resource created for one tenant does not automatically belong to another tenant.

14. Security Considerations

Phase 3 introduces limited security requirements.

The primary security rule is:

Only authenticated users may create tenants.

Ownership Protection

Tenant update and deactivation operations should be restricted to the tenant's initial owner reference until membership-based authorization exists.

Backend Enforcement

Tenant operations must be validated by the backend.

Frontend controls must never be considered security controls.

Frontend filtering is not a valid authorization mechanism.

Authorization Boundary

Phase 3 does not implement final tenant authorization.

Final tenant authorization will be membership-based in later phases.

15. High-Level Persistence Design

The tenant should be stored as a persistent business entity.

At a high level, the tenant record should include:

- Unique identifier
- Business name
- Business type
- Timezone
- Tenant status
- Initial owner reference
- Creation timestamp
- Update timestamp

This document intentionally does not define:

- SQL schemas
- ORM models
- Prisma schemas
- TypeORM entities
- Migrations
- Database indexes

Those belong to implementation design.

16. Testing Strategy

Testing should focus exclusively on tenant behavior.

Tenant Creation Tests

Verify:

- Authenticated users can create tenants.
 - Unauthenticated users cannot create tenants.
 - New tenants start as PENDING_ONBOARDING.
 - Owner reference is recorded.
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Tenant Validation Tests

Verify:

- Required business name rules are enforced.
 - Invalid metadata is rejected.
 - Required tenant information is validated.
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Tenant Update Tests

Verify:

- Tenant metadata can be updated.
 - Owner reference remains unchanged.
 - Updates do not create future-phase entities.
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Tenant Deactivation Tests

Verify:

- Tenant can be deactivated.
 - Tenant record remains preserved.
 - Deactivated tenants are not treated as active.
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Scope Tests

Verify that Phase 3 does not implement:

- Memberships
- Roles
- Services

- Staff
 - Appointments
 - Branding
 - Notifications
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17. Acceptance Criteria

Phase 3 is complete when:

1. Tenant exists as a persistent business entity.
 2. Authenticated users can create tenants.
 3. New tenants start in PENDING_ONBOARDING.
 4. Initial owner reference is stored.
 5. Tenant metadata can be updated safely.
 6. Tenant can be deactivated.
 7. Tenant records are preserved.
 8. Tests validate tenant behavior.
 9. Future-phase functionality remains unimplemented.
 10. Documentation is complete.
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18. Risks

Risk: Membership Functionality Introduced Too Early

Mitigation:

Do not implement memberships in Phase 3.

Risk: Owner Reference Becomes Permanent Authorization

Mitigation:

Ownership transition rule explicitly requires future membership-based authorization.

Risk: Branding Introduced Prematurely

Mitigation:

Branding is out of scope.

Risk: Lifecycle Complexity

Mitigation:

Restrict lifecycle to:

- PENDING_ONBOARDING
- ACTIVE
- DEACTIVATED

Risk: Weak Ownership Validation

Mitigation:

Restrict tenant modifications to the initial owner reference until membership-based authorization exists.

19. Review Checklist

Verify:

Scope

- Tenant model only.
- No memberships.
- No roles.
- No services.
- No appointments.
- No branding.
- No notifications.

Tenant Creation

- Authentication required.
- Owner reference created.
- Default lifecycle state is correct.

Tenant Lifecycle

- Only approved lifecycle states exist.

- Deactivation preserves tenant records.
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Security

- Backend validation exists.
 - Frontend filtering is not trusted.
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Documentation

- Ownership transition rule documented.
 - Future-phase boundaries documented.
 - Tenant ownership principles documented.
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20. Stop Condition

Phase 3 ends when:

- Tenant model is implemented.
- Validation passes.
- Tests pass.
- Documentation is complete.
- Review checklist passes.

After Phase 3:

- Do not implement memberships.
- Do not implement roles.
- Do not implement permissions.
- Do not implement services.
- Do not implement appointments.

The project must stop and wait for explicit approval before Phase 4 is designed or implemented.

Accepted approval examples:

- Approved
- Continue
- Proceed to Phase 4

Any other message must not be interpreted as approval.

End of Document

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